Asynchronous Communications (Kafka)

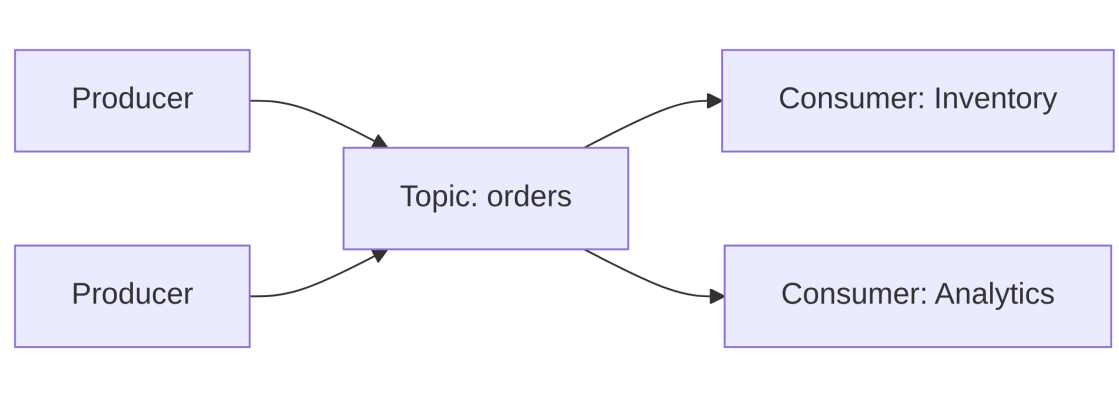
Apache Kafka is a **distributed streaming platform** designed for **high-throughput, fault-tolerant, and scalable** data pipelines. Kafka enables applications to:

- **publish and subscribe to streams of records**
- **store records durably**
- **process streams in real-time**

Key Components

- **Broker**: Kafka server that stores records and serves clients.
- **Producer**: Application that publishes messages to Kafka topics.
- **Consumer**: Application that subscribes to topics and processes messages.

Kafka Topics


Producer
Producer
Topic: orders
Consumer: Inventory
Consumer: Analytics

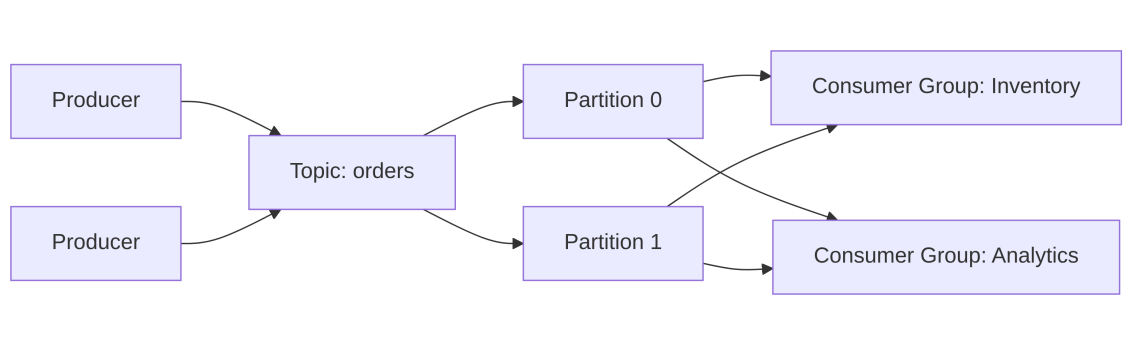
A **topic** is like a **channel** or **folder** where messages (called *records*) are stored. Producers write to a topic, consumers read from it. Topics enable **logical separation** of data streams.

- A topic is **append-only** (Kafka never modifies existing events).
- Consumers can read the same topic independently without interfering with each other.
- **Offset**: Each message in a partition has a unique sequential ID called an **offset**.
- **Retention**: Kafka keeps messages for a configurable duration or size, independent of consumption.
- **Replay**: Consumers can re-read messages by resetting their offsets.

Example topics:

- `orders`
- `payments`
- `temperature-readings`

Kafka Partitions


Producer
Producer
Topic: orders
Partition 0
Partition 1
Consumer Group: Inventory
Consumer Group: Analytics

A **partition** is the *unit of parallelism and scaling* inside a topic.

A topic is split into N partitions:

```
Topic: orders
    ├── Partition 0
    ├── Partition 1
    └── Partition 2
```

```
Partition 0: [event1][event2][event3]...
Partition 1: [event4][event5][event6]...
Partition 2: [event7][event8]...
```

A producer decides **which partition** to send a message to:

- By key hashing (preferred)
- Round-robin (if no key)
- Custom partitioner

Scalability

Partitions allow Kafka to scale horizontally across many brokers.

More partitions → More consumers can process the topic **in parallel** → higher throughput.

Replication

Partitions can be **replicated** across brokers for durability and fault tolerance.

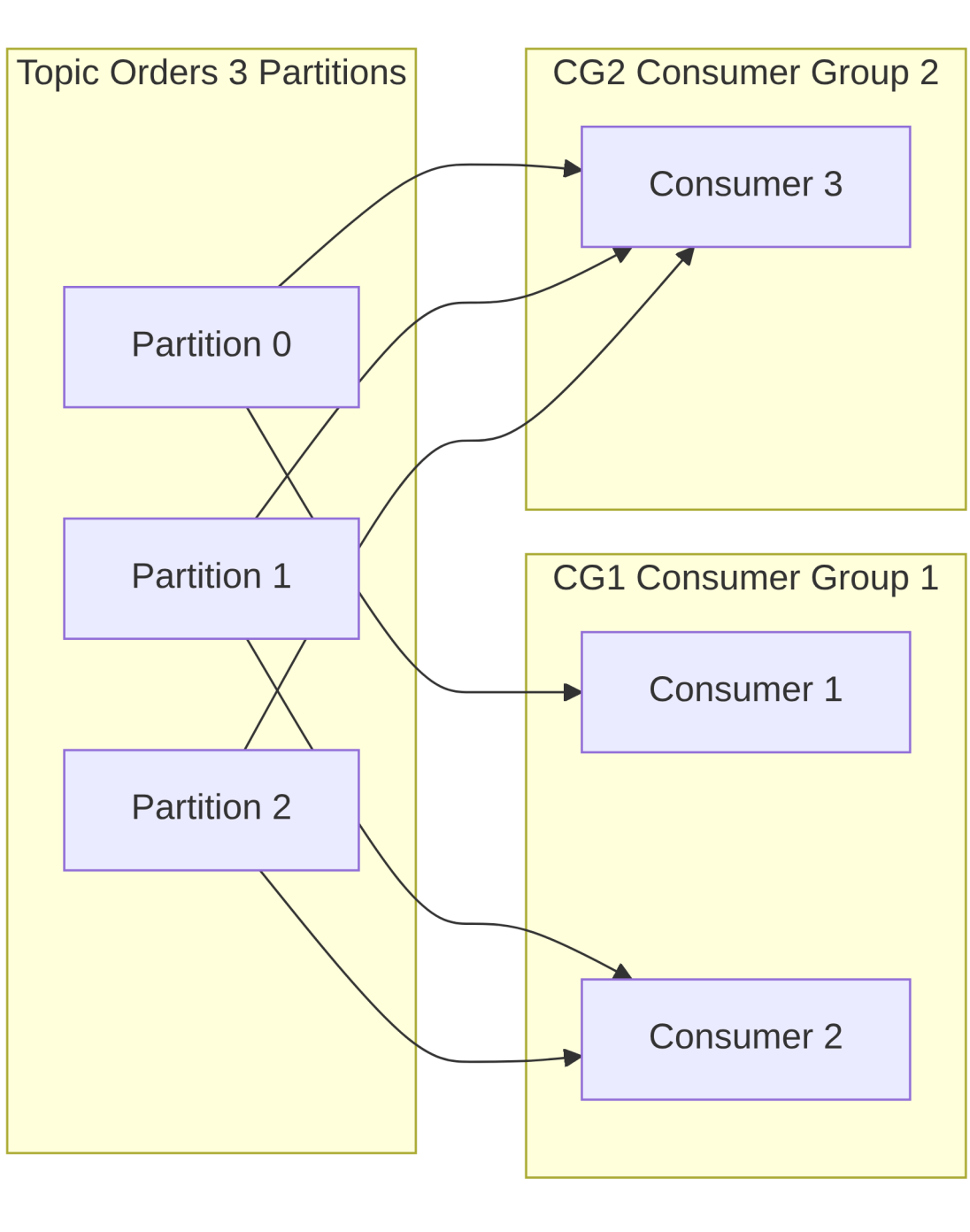
Example:

```
Partition 0:
  Leader: Broker 1
  Followers: Broker 2, Broker 3
```

If the leader fails, a follower becomes the new leader (**RAFT**).

Kafka – Consumer Groups

- **Consumer Group** = a set of consumers that **work together to read a topic**
- **Each partition** is read by **only one consumer per group** → preserves **message order**
- **Multiple consumers in the same group** → Kafka **balances partitions** among them
- **Multiple consumer groups** → each group receives **all messages independently**


Topic Orders 3 Partitions
Partition 0
Partition 1
Partition 2
CG2 Consumer Group 2
Consumer 3
CG1 Consumer Group 1
Consumer 1
Consumer 2

Ordering

Ordering is guaranteed only within one partition. Across partitions, there is *no ordering guarantee*.

If you need ordering per key (e.g., same `orderId`), you send messages using a **key**, and Kafka ensures: **same key → same partition → same order**.

Resources

- Kafka Official Documentation
- Kafka Tutorials
- Kafka in Action
- https://www.redpanda.com/